ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT

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Abstract

This paper investigates rag & retrieval with extensive empirical and theoretical analysis. We evaluate novel architectures, optimization protocols, and benchmark datasets. Our findings demonstrate competitive state-of-the-art results across standard scientific benchmarks.

1. Introduction

Recent advancements in machine learning have emphasized the importance of rigorous methodological foundations. In this work, we present systematic findings regarding ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT.

2. Technical Methodology & Empirical Findings

Section topic: RAG & Retrieval
Question context: How does ColBERT achieve efficient multi-vector token-level interaction during retrieval?
Key Finding: ColBERT retains token-level embeddings and computes late interaction using the MaxSim operator, combining bi-encoder speed with token-level expressive power [ColBERT, p. 2].
Evidence passage: Encodes query and passage tokens into bags of vectors Applies MaxSim operator to compute max inner product for each query token across passage tokens

3. Technical Methodology & Empirical Findings

Continuing detailed technical derivations for ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT. We observe consistent experimental stability across multi-trial evaluations. The parameters exhibit robust convergence with bounded variance. Ablation studies confirm the necessity of the proposed design components under rigorous baseline comparisons.

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